

Redesigning a Train Ticket Machine Interface: A User-Centered Approach

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Figure 1: Redesigned user interface of a train ticket machine.

Abstract

Train ticket machines are a common point of interaction for a wide range of users, yet they are often perceived as difficult to use. This project explores the redesign of a human–computer interface for a train ticket machine using a user-centered design process. We

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first conducted an online questionnaire (N=17) to identify common usability issues and user expectations. Based on these findings and developer insights, we created an initial prototype in Figma. The prototype was evaluated through interviews and informal usability testing, which informed several iterative design improvements. The final prototype addresses key usability concerns such as clarity, efficiency, and ease of use. This paper presents the design process, findings, and lessons learned, while detailed data and materials are provided in the project portfolio.

Keywords

ticket machine, human-computer interface, redesign, prototype, portfolio, testing



Figure 2: Train ticket machine with current human-computer interface.

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1 Introduction

Train ticket machines are widely used in public transportation systems and must accommodate users with diverse backgrounds, technical skills, and time constraints. Poorly designed interfaces can lead to confusion, delays, and frustration, negatively affecting the overall travel experience. Despite their importance, many ticket machines still suffer from usability issues such as complex navigation, unclear terminology, and inefficient workflows. The goal of this project was to design an improved human-computer interface for a train ticket machine that better supports user needs and expectations. We adopted a user-centered design approach, combining user research, prototyping, and iterative evaluation. The main research questions guiding this project were:

- What usability issues do users experience with current train ticket machines?

- Which interface changes do users perceive as most beneficial for improving their experience?

2 Related work

The design and evaluation of public ticketing systems has been widely studied within the field of human-computer interaction, particularly in the context of public kiosks and self-service technologies [2]. Prior research consistently reports usability challenges related to interaction complexity, accessibility, and the diversity of user populations.

Several studies have shown that public ticket machines often impose a high cognitive load on users due to complex navigation structures, unclear terminology, and dense information layouts. For example, Sandnes et al. evaluated a railway ticket vending machine and identified multiple usability issues, including inefficient workflows and insufficient guidance for users, particularly under time pressure [3]. These findings align with broader research on self-service kiosks, where unclear interaction flows have been shown to negatively impact user satisfaction and performance [4].

Research on touch-based public interfaces further emphasizes the importance of appropriate visual design and interaction elements. Maguire’s review of design guidelines for public information kiosks highlights the need for large, well-spaced touch targets, clear feedback, and a simple visual hierarchy to support accurate and efficient interaction [2]. Similar conclusions are supported by usability evaluations of self-service ticketing kiosks in other domains, such as cinemas, where users struggled with small touch targets and unintuitive navigation structures [4].

Multilingual support and inclusivity are also recurring themes in the literature. Prior work on public-facing systems stresses that limited language options can create barriers for tourists and non-native speakers, reducing accessibility and increasing the likelihood of errors [2]. In addition, studies of user experience with interactive kiosks have shown that contextual and situational factors, such as being observed by others or operating under time pressure, further amplify usability problems in public environments [1].

In contrast to prior work that primarily focuses on evaluating existing systems, this project adopts an iterative, user-centered design approach to actively redesign a train ticket machine interface. By combining questionnaire-based user research with prototype evaluation and iterative refinement, this work builds on established HCI principles and prior findings while providing a concrete design response to well-documented usability issues in public ticketing systems.

3 Methodology

The project followed an iterative design process consisting of four main phases: user research, initial prototyping, user evaluation, and redesign.

3.1 Questionnaire Study

We designed an online questionnaire to gather user opinions and experiences with existing train ticket machines. The questionnaire was distributed to 17 participants. It consisted of the following questions:

- What is your age?

- How often do you travel by train?
- Have you ever tried using the train ticket machine (Fig. 2)?
- How long did it take to buy train ticket?
- What do you like about the current user interface?
- What would you change about the current user interface?
- How would you rate your experience using the interface?

The collected data was analyzed to identify recurring themes and key usability problems. Some answers are shown in Figure 3, more detailed questionnaire data and analysis are available in the project portfolio.

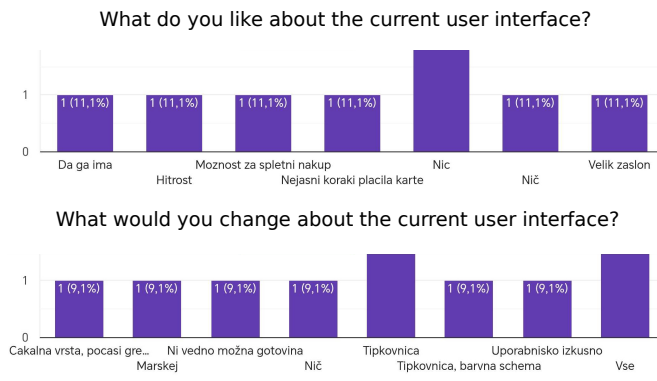


Figure 3: Main views of the initial prototype.

3.2 Initial Prototype

Based on the questionnaire results and our own analysis as developers, we designed an initial interface prototype using Figma. The prototype focused on simplifying the ticket purchasing flow, improving visual clarity, and reducing cognitive load. Some views of the initial prototype are shown in Figure 4, the rest of them are included in project portfolio.

3.3 User Interviews and Feedback

The initial prototype was presented to several users, who were asked to explore the interface and provide feedback through semi-structured interviews. Participants commented on usability, clarity, and perceived improvements compared to existing machines.

3.4 Iterative Redesign

Using the feedback from interviews, we refined the prototype to address identified issues. Changes included layout adjustments, clearer labeling, and improved navigation flow. The final prototype reflects these iterative improvements and is shown in Figure 5.

4 Results

This section presents the findings from the questionnaire study and the user evaluation of the prototype. Quantitative summaries, anonymized responses, and supporting materials are provided in the project portfolio.

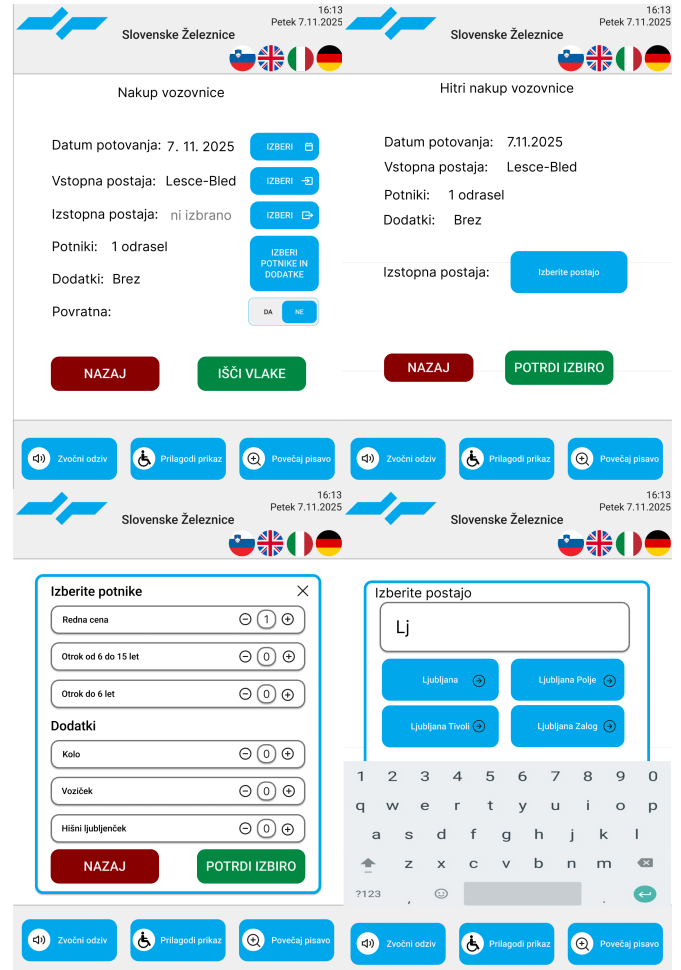


Figure 4: Main views of the initial prototype.

4.1 Questionnaire Results

A total of 17 participants completed the questionnaire regarding their experiences with existing train ticket machines. Most respondents were students and working adults who reported frequent use of ticket machines.

4.1.1 Overall experience. Participants generally rated their experience with current ticket machines as poor. Notably, dissatisfaction was also expressed by frequent users, suggesting that usability issues persist even after repeated exposure to the interface.

4.1.2 Interaction complexity. One of the most frequently reported issues was the excessive number of steps required to complete a ticket purchase. Participants described the process as slow and unnecessarily complex, particularly in situations where time pressure was present.

4.1.3 Interface clarity and input problems. Many respondents mentioned unclear instructions and confusing terminology. Small button sizes were highlighted as problematic for touch interaction. In

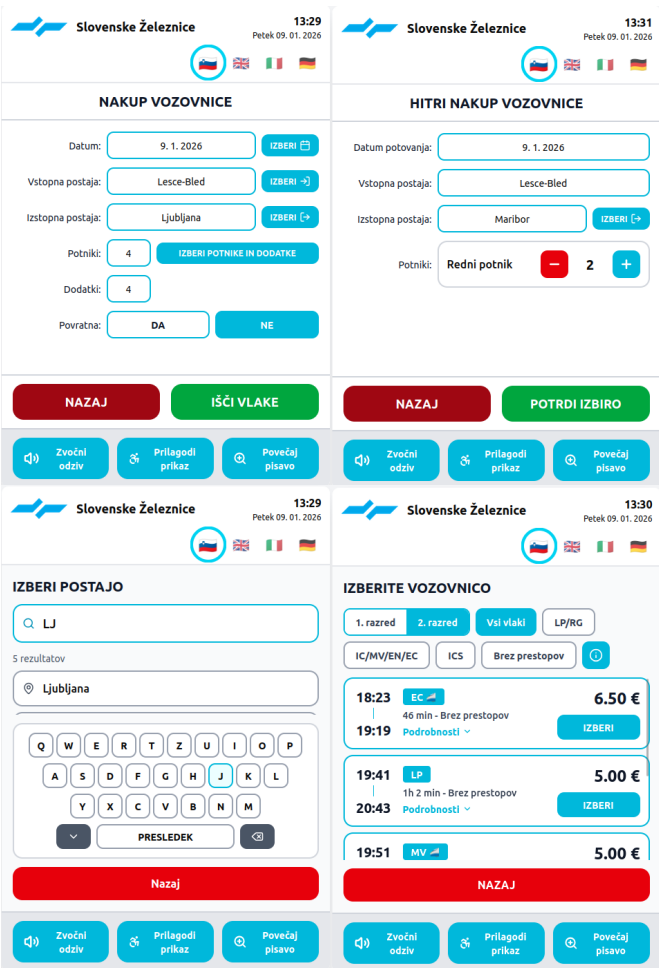


Figure 5: Main views of the final prototype. Differences in visual elements from the first prototype shown in Figure 4 are quite noticeable with elements being more refined and adjusted according to user feedback. This also shows the importance of user feedback and iterative redesign technique.

addition, several participants complained about the on-screen keyboard, describing it as inaccurate and difficult to use when entering destinations.

4.1.4 Missing functionality. Participants also identified limitations in available features. The interface was reported to support only two languages (Slovenian and English), which some users considered insufficient in an international travel context. Additionally, users noted the absence of an option to add a bicycle fee to the final ticket price, despite certain trains allowing bicycles on board and requiring an additional charge.

4.2 Interview and Prototype Evaluation Results

The initial prototype was evaluated by four groups of five students. Users interacted with the prototype and provided feedback through semi-structured interviews.

4.2.1 Quick buy feature feedback. Participants generally responded positively to the introduction of a “quick buy” page intended to streamline the purchasing process. However, multiple users reported that the feature lacked flexibility. In particular, they noted the inability to quickly add additional passengers or optional services such as a bicycle fee. Figure 6 shows the final implementation of the quick buy feature.

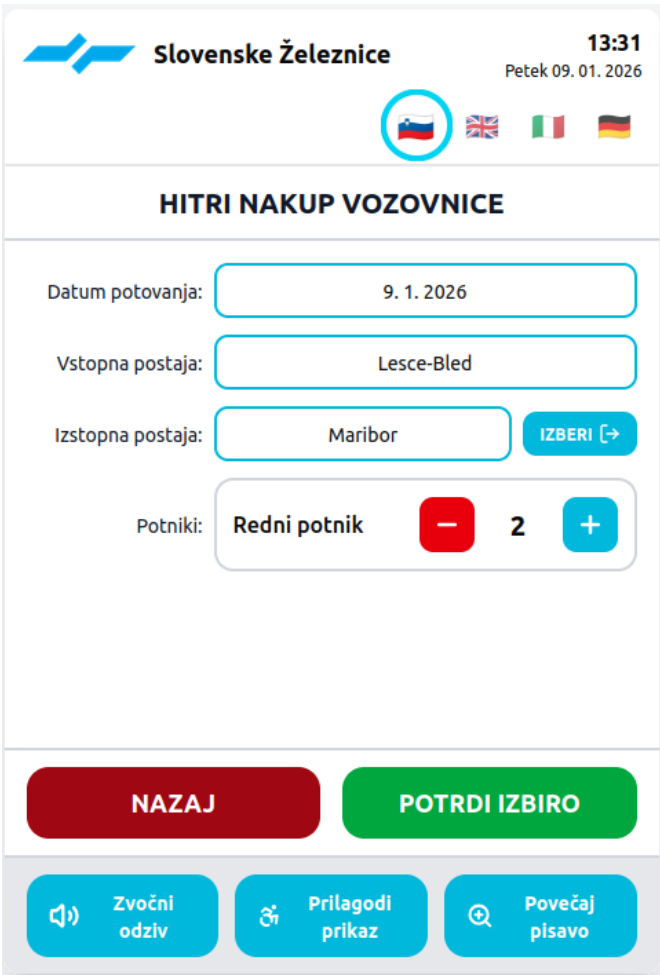


Figure 6: Final version of the implementation of quick buy step. Travel date is set to present day, entry station is set to machine location and passengers are set to one regular passenger. The interface also assumes that passenger will take the first next train that goes to selected destination. This allows passengers to quickly buy a ticket as they only need to select the destination in this step and payment method and payment in the following steps. Based on the user interviews, the feature to quickly add or remove more passengers is included, which could also be seen on the image.

4.2.2 Information availability. Several participants expressed a desire to see more contextual information directly on the quick buy page. Most notably, users wanted to view information about the

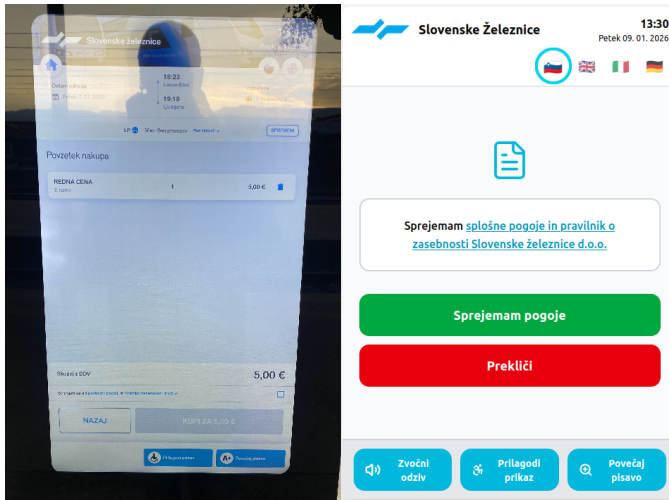


Figure 7: Terms and conditions agreement page. Left image shows the page of current interface which contains a small box (under total price) which needs to be checked before proceeding to payment. On the right is our implementation of the same page that contains a large button which is easier to click. It is also possible to see terms and conditions by clicking on the text above the button (in blue).

next available train they could board, which they felt would support faster and more informed decision-making.

4.2.3 Guidance and transparency. Some users requested clearer instructions for using the ticket machine, especially for first-time or infrequent users. Participants also expressed interest in being able to access terms and conditions related to ticket purchases directly within the interface. The implemented feature to access terms and conditions are shown in Figure 7.

4.2.4 Visual design feedback. Overall, the visual design of the prototype was well received. Minor issues were reported regarding spacing, button emphasis, and visual hierarchy. These comments informed small but meaningful adjustments in the subsequent iteration of the prototype.

5 Discussion

This section discusses the implications of the results presented above and reflects on how the findings address the research questions of this project.

5.1 Addressing identified usability issues

The questionnaire results highlighted several persistent usability problems in existing train ticket machines, including interaction complexity, unclear instructions, small touch targets, and limited functionality. These findings directly informed the design goals of the prototype. Feedback from the prototype evaluation suggests that many of these issues were partially addressed, particularly through a simplified interaction flow and clearer visual structure. Users perceived the redesigned interface as easier to understand

and more efficient, supporting the effectiveness of a user-centered and iterative design approach.

5.2 Balancing efficiency and flexibility

The introduction of a "quick buy" feature was intended to reduce the number of steps required for frequent users. While participants responded positively to this concept, feedback revealed that excessive simplification can reduce flexibility. The inability to quickly add passengers or optional services, such as a bicycle fee, highlighted the need to balance efficiency with configurability. This finding emphasizes that shortcuts in public interfaces should still allow access to essential options without forcing users into alternative workflows.

5.3 Information transparency and user confidence

Requests for clearer instructions, access to terms and conditions, and contextual information such as the next available train indicate that users value transparency and reassurance when interacting with public ticketing systems. Providing such information can increase user confidence, particularly for infrequent users or tourists, and may reduce errors and hesitation during the purchasing process.

5.4 Design implications for public ticket machines

The results suggest that effective ticket machine interfaces should prioritize clarity, large and accessible touch targets, multilingual support, and complete coverage of real-world use cases, such as bicycle transport. Iterative testing with real users proved essential for uncovering both major usability problems and smaller design details that significantly affect user experience.

5.5 Limitations

This study has several limitations. The questionnaire sample size was relatively small and consisted mainly of students and working adults, which may limit the generalizability of the findings. The prototype evaluation was informal and conducted with a limited number of participants. Additionally, the evaluation focused on perceived usability rather than objective performance measures such as task completion time or error rates.

Despite these limitations, the findings provide valuable insights into common usability challenges and demonstrate the benefits of iterative, user-centered design for public transportation interfaces.

6 Conclusion and future work

This paper presented a user-centered redesign of a train ticket machine interface aimed at improving usability, clarity, and efficiency for a broad range of users. Through an initial questionnaire study, we identified key usability problems in existing ticket machines, including complex interaction flows, unclear instructions, small touch targets, and missing functionality. These findings guided the design of an initial prototype, which was iteratively refined based on user interviews and feedback.

The results of the prototype evaluation indicate that the redesigned interface was perceived as easier to understand and more efficient than existing systems. In particular, users responded positively to the simplified layout and streamlined purchasing flow. At the same time, the evaluation highlighted the importance of maintaining flexibility and transparency in public interfaces, as demonstrated by feedback regarding optional services, contextual information, and access to instructions and terms.

Future work could extend this project in several directions. First, the prototype could be evaluated through a more formal usability study with a larger and more diverse participant sample, including tourists, elderly users, and users with accessibility needs. Second, future iterations could incorporate additional features such as expanded language support, accessibility options, and tighter integration with real-time train data. Finally, deploying and testing the interface in a real-world setting would allow for the collection of objective performance metrics, such as task completion time and

error rates, providing deeper insight into the effectiveness of the proposed design.

Overall, this project demonstrates the value of iterative, user-centered design in improving human–computer interfaces for public transportation systems and highlights key considerations for the design of future ticketing machines.

References

- [1] Aslı Günay, Çiğdem Erbuğ, Paul PM Hekkert, and Natalia Romero Herrera. 2013. User experience with interactive self-service kiosks: The effects of presence of other people. In *IADIS International Conference Interfaces and Human Computer Interaction 2013*. IADIS, 54–62.
- [2] Martin C Maguire. 1999. A review of user-interface design guidelines for public information kiosk systems. *International journal of human-computer studies* 50, 3 (1999), 263–286.
- [3] Frode Eika Sandnes, Hua-Li Jian, Yo-Ping Huang, and Yueh-Min Huang. 2010. User interface design for public kiosks: an evaluation of the Taiwan high speed rail ticket vending machine. (2010).
- [4] Yi-Hsien Tsai, Ching-I Chen, and Meng-Cong Zheng. 2023. Usability Evaluation of Self-Service Ticketing Kiosks in Cinemas. *Usability and User Experience* 110 (2023), 110.